

HARDIK SONU

Materials Engineering Student | Semiconductor Process & Defect Analysis | Computational Materials / AI

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PROFESSIONAL SUMMARY

Second-year Metallurgical and Materials Engineering student combining materials characterization and selection fundamentals with applied computational skills (Python, MATLAB). Building PackageGuard AI, an in-development semiconductor defect-detection tool aimed at reducing manual inspection time. Completed Materials Data Science and Informatics training (Georgia Tech). Seeking a semiconductor process engineering or materials engineering internship.

EDUCATION

BS Metallurgical and Materials Engineering

2025 – Present

Institute of Metallurgy and Materials Engineering, University of the Punjab, Lahore

- Current GPA: 3.5+/4.0 | Relevant coursework: Materials Characterization, Materials Selection, Metallurgy, QHSE Principles, Semiconductor Materials Fundamentals, Data-Driven Engineering (MATLAB, Python)

Intermediate (F.Sc., Pre-Engineering)

2023 – 2025

Iqra Postgraduate College, Sadiqabad — Grade: A+

Matriculation

2021 – 2023

Govt. Tameer-e-Millat High School, Sadiqabad

RELEVANT EXPERIENCE

Virtual Intern — Pio Petro (Oil & Gas)

Aug 2026 – Sep 2026

- Completing a structured virtual internship examining the role of a metallurgical and materials engineer within the oil and gas sector, including materials selection and integrity fundamentals.
- Studying applied petroleum and gas engineering principles as they relate to material performance and process reliability, building foundational process-engineering exposure ahead of core coursework.

ADDITIONAL EXPERIENCE

Presentation Designer (Internship) — Webgross

Jan 2026 – Feb 2026

Tech & Branding Solutions Provider — New Delhi, India (Remote)

- Designed 15+ client-facing PowerPoint presentations and branding materials, translating technical and business content into clear, professional visuals for cross-industry clients.

Freelancer — Fiverr

Aug 2025 – Present

- Deliver freelance services in document formatting, CV design, and presentation creation for international clients; completed 20+ projects on schedule with a 100% positive feedback record.
- Strengthened client communication and independent project/time-management skills through end-to-end freelance delivery.

LEADERSHIP & VOLUNTEER EXPERIENCE

Designer Member & Executive Member — Science Communication Society (SciComS), IMME, University of the Punjab

2025 – Present

- Developed and deployed an automated certificate-distribution program in Python that identifies webinar attendees from uploaded name/email lists and issues personalized certificates to 300+ recipients simultaneously, eliminating manual repetitive work for the society.

Volunteer — Foreign Edge Consultants

2025

- Supported operations at Pakistan's largest Study Expo, assisting 500+ attendees and managing visitor registration.

PROJECTS

PackageGuard AI — *Flagship Project (In Development)*

- Building an AI-based semiconductor defect detection application in Python to address a recurring process problem: manual inspection and testing of devices for defects is slow and costly.
- Designing the app to classify defect type and generate a confidence/quality score, giving process engineers a fast go/no-go signal on whether a unit should proceed to the next process step.
- Result: functional working prototype completed, with public release planned; targets reduced physical testing time and cost by flagging likely defects before full manual inspection.

Automated Certificate & Email Distribution System — *SciComS*

- Built with Python to solve a process-efficiency problem: manually generating and emailing individual webinar certificates did not scale.
- Automated attendee identification from uploaded name/email lists and batch-generated personalized certificates for direct email dispatch.
- Result: 300+ certificates distributed simultaneously, eliminating hundreds of manual emails and freeing up volunteer time.

Company Dashboard

- Built a data dashboard to solve a visibility problem for a local company: key business metrics were not centrally tracked.
- Designed data visualization views to surface trends and metrics in an accessible format for non-technical stakeholders.
- Result: gave the company a single reference point for tracking key business metrics, replacing ad hoc manual reporting.

Additional Python Projects

- Hand-Gesture Controlled Hill Climb Game — real-time computer-vision hand-tracking mapped to in-game controls; fully playable, demonstrating applied CV and signal-processing skills.
- Jarvis (Personal AI Assistant) — NLP-based command/response system for reminders and task lookups, built as a self-directed applied-AI project.
- University of the Punjab Merit Calculator — automated the university's merit formula, reducing calculation errors and turnaround time for applicants.

CERTIFICATIONS & ONLINE COURSES

- Materials Data Sciences and Informatics — Georgia Institute of Technology (Coursera), Jul 2026
- Professional Certificate in Health and Safety Engineering — Khalifa University (Coursera), Jun 2026
- Certificate of Appreciation for Research Presentation, “How Microstructure Controls the Properties of a Material” — Lahore College for Women University, Feb 2026
- MATLAB Basics — Vanderbilt University (Coursera)
- Crash Course on Python Programming — Google (Coursera), Aug 2026

SKILLS

Materials Engineering: Materials Characterization (foundational coursework), Materials Selection, Materials Science Fundamentals, Metallurgy, Semiconductor Materials Fundamentals, QHSE (Quality, Health, Safety, and Environment), Failure Analysis (foundational)

Semiconductor & Process: Semiconductor Materials Fundamentals, Defect Detection & Classification (applied — PackageGuard AI), Materials Data Sciences & Informatics (Georgia Tech, Coursera), Process Analysis

Computational / Data: Python, MATLAB (Intermediate), Data Analytics, Process Automation, C++ (Basics), Computer Vision (applied/project level), Natural Language Processing (applied/project level)

Software & Office Tools: MS Word, Microsoft Excel, Microsoft PowerPoint, Google Sheets, FreeCAD (Intermediate), Document Formatting

Communication & Design: Presentation Design, Public Speaking, Technical Documentation, Cross-Functional Collaboration

LANGUAGES

English | Urdu | Punjabi | Other Local Languages